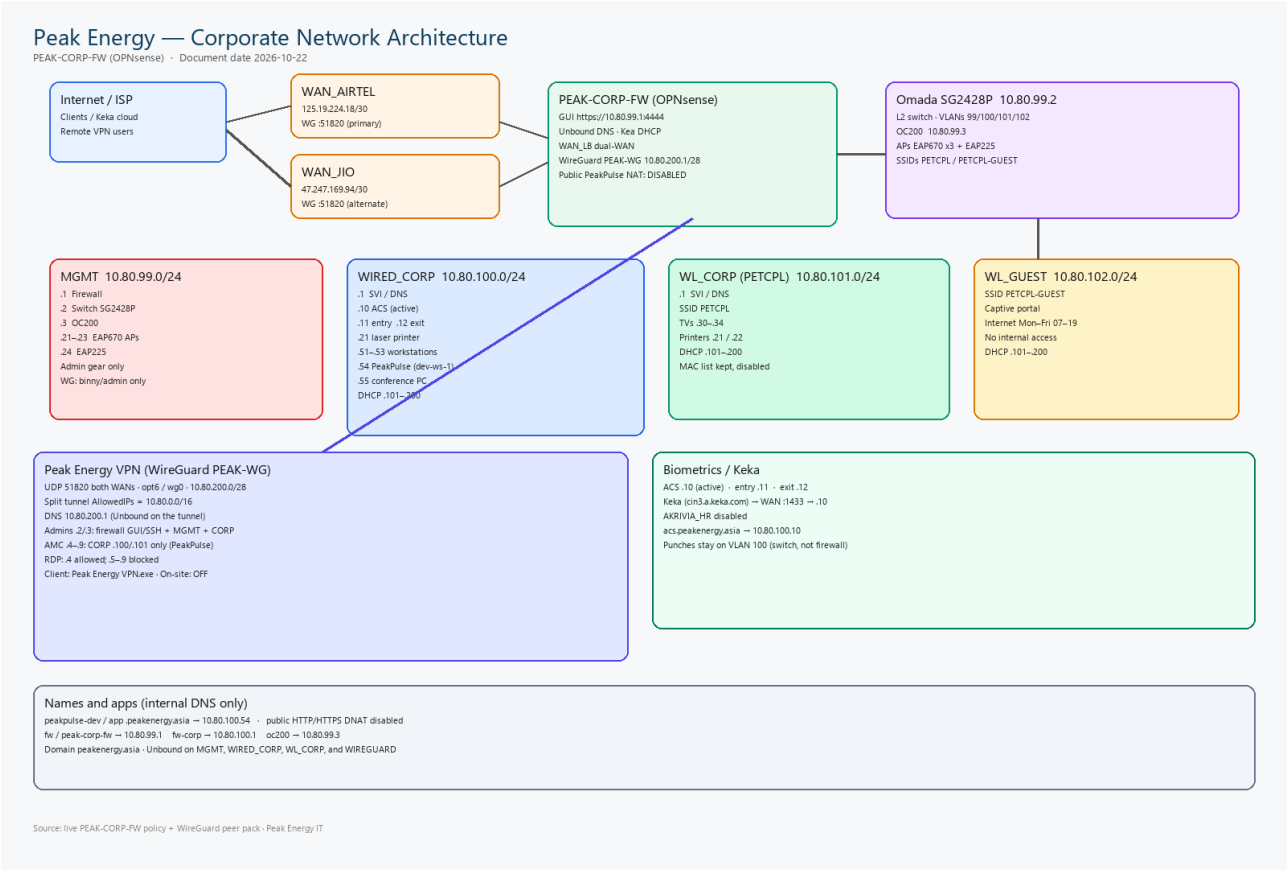
Peak Energy — Corporate Network Architecture

Firewall: **PEAK-CORP-FW** (OPNsense) · Generated 2026-10-22

This document describes the live network: VLANs, WAN links, Omada Wi■Fi, biometrics and Keka, WireGuard (Peak Energy VPN), and access policy for PeakPulse.



1. WAN links

WAN	Public IP	Gateway name	Notes
Airtel Fiber	125.19.224.18/30	WAN_AIRTEL	Primary; WG endpoint default
Jio RF	47.247.169.94/30	WAN_JIO	Failover / alternate WG endpoint
WAN_LB	—	WAN_LB	Dual-WAN load balance + failover

2. VLANs / subnets

VLAN / Role	Subnet	Gateway (FW)	DHCP pool	Interface
MGMT (admin gear)	10.80.99.0/24	10.80.99.1	10.80.99.101–200	lan / re0
WIRED_CORP	10.80.100.0/24	10.80.100.1	10.80.100.101–200	opt2 / VLAN 100
WL_CORP (PETCPL)	10.80.101.0/24	10.80.101.1	10.80.101.101–200	opt3 / VLAN 101
WL_GUEST	10.80.102.0/24	10.80.102.1	10.80.102.101–200	opt4 / VLAN 102
WireGuard VPN	10.80.200.0/28	10.80.200.1	static peers	opt6 / wg0

3. MGMT hosts (10.80.99.0/24)

IP	Hostname	Role
10.80.99.1	PEAK-CORP-FW / OpnSense	Firewall GUI :4444, SSH, Unbound DNS
10.80.99.2	sg2428p	Omada switch SG2428P
10.80.99.3	oc200	Omada OC200 controller https://10.80.99.3/
10.80.99.21	eap670-1	Wi-Fi AP EAP670
10.80.99.22	eap670-2	Wi-Fi AP EAP670
10.80.99.23	eap670-3	Wi-Fi AP EAP670
10.80.99.24	eap225-3	Wi-Fi AP EAP225

4. WIRED_CORP hosts (10.80.100.0/24)

IP	Hostname	Role
10.80.100.1	—	WIRED_CORP SVI / DNS
10.80.100.10	biometric-server	ACS / Attendance SQL
10.80.100.11	biometric-entry	Biometric entry terminal
10.80.100.12	biometric-exit	Biometric exit terminal
10.80.100.21	laser-printer	HP LaserJet (Ethernet)
10.80.100.51	peak-ws-1	Workstation 1
10.80.100.52	peak-ws-2	Workstation 2
10.80.100.53	peak-ws-3	Workstation 3
10.80.100.54	dev-ws-1	PeakPulse / peakpulse-dev.peakenergy.asia
10.80.100.55	peak801-in-conf	Conference room PC

5. WL_CORP hosts (10.80.101.0/24)

IP	Hostname	Role
10.80.101.1	—	WL_CORP SVI / DNS
10.80.101.21	laser-printer	HP LaserJet (Wi-Fi)
10.80.101.22	printer-md	MD HP Smart Tank Wi-Fi
10.80.101.30	tv-conferance	TV Conference Room
10.80.101.31	tv-kalaam	TV Abdul Kalaam
10.80.101.32	tv-ratan	TV Ratan Tata

10.80.101.33	tv-onm	TV Operations & Mgmt
10.80.101.34	tv-md	TV MD

6. WireGuard / Peak Energy VPN

The Windows client is **Peak Energy VPN**. On PETCPL or wired corp the tunnel must be off, because AllowedIPs sends 10.80.0.0/16 into WireGuard.

Peer	Tunnel IP	Access
PEAK-WG server	10.80.200.1/28	OPNsense wg0 listen UDP 51820
binny.andrews	10.80.200.2/32	Admin: FW GUI/SSH + MGMT + CORP
admin	10.80.200.3/32	Admin: FW GUI/SSH + MGMT + CORP
jagadeshwar	10.80.200.4/32	AMC: CORP .100/.101; RDP allowed
venu.gopal.reddy	10.80.200.5/32	AMC: CORP only; RDP blocked
poovarasu	10.80.200.6/32	AMC: CORP only; RDP blocked
sanoj.james	10.80.200.7/32	AMC: CORP only; RDP blocked
sampada	10.80.200.8/32	AMC: CORP only; RDP blocked
sharan	10.80.200.9/32	AMC: CORP only; RDP blocked

7. Biometrics and Keka

Attendance stays on WIRED_CORP. Entry and exit terminals talk to the ACS on the same VLAN, so that traffic does not pass the firewall. Keka cloud pulls SQL from the public WAN address on TCP 1433. NAT on Airtel and Jio forwards that only from KEKA_HR to 10.80.100.10, the active ACS. The old Akrivia source alias is kept and disabled.

Host	Address	Role today
biometric-server	10.80.100.10	Active ACS. Alias ACS_SERVER. Keka SQL target.
biometric-entry	10.80.100.11	Entry terminal. Same VLAN as ACS.
biometric-exit	10.80.100.12	Exit terminal. Same VLAN as ACS.

8. Internal DNS (Unbound)

Name	Address	Use
fw.peakenergy.asia	10.80.99.1	Firewall (MGMT)
peak-corp-fw.peakenergy.asia	10.80.99.1	Firewall (MGMT)
fw-corp.peakenergy.asia	10.80.100.1	Firewall (WIRED_CORP)
acs.peakenergy.asia	10.80.100.10	Attendance / ACS SQL
oc200.peakenergy.asia	10.80.99.3	Omada controller
app.peakenergy.asia	10.80.100.54	PeakPulse
peakpulse-dev.peakenergy.asia	10.80.100.54	PeakPulse

9. Security & access policy

Topic	Current policy
PeakPulse public WAN	DISABLED — HTTP/HTTPS DNAT kept but turned off on both WANs
PeakPulse access	On-site LAN or WireGuard only → 10.80.100.54 (dev-ws-1)
WG DNS	New peers use 10.80.200.1 (Unbound on WIREGUARD). Legacy may use 10.80.100.1. Both are allowed.
WG AllowedIPs	10.80.0.0/16 split tunnel (corp only, not full internet)
WG endpoints	125.19.224.18:51820 (Airtel, default) or 47.247.169.94:51820 (Jio)
WG interface	opt6 / wg0, description WIREGUARD. Routes disabled; firewall does the policy.
MGMT via VPN	Only WG_FW_ADMINS (binny.andrews .2 and admin .3)
FW GUI / SSH via VPN	Only WG_FW_ADMINS. GUI listens on MGMT, WIRED, WL_CORP, and WIREGUARD — never WAN.
AMC via VPN	CORP only: 10.80.100.0/24 and 10.80.101.0/24. No MGMT, no firewall GUI.
RDP over VPN	TCP 3389 blocked for Venu, Poovarasu, Sanoj, Sampada, Sharan. Allowed for binny, admin, Jagadeshwar.

ACS SQL from Internet	KEKA_HR only (cin3.a.keka.com) → 10.80.100.10:1433 on both WANs
Akrivia	AKRIVIA_HR alias kept, disabled
Corp Wi-Fi MAC list	WL_CORP_MACS rules exist but are DISABLED. PETCPL is open to the VLAN.
Guest Wi-Fi	SSID PETCPL-GUEST, VLAN 102, captive portal. Internet Mon–Fri 07:00–19:00. No internal access.
On-site vs remote	On PETCPL or wired corp, Peak Energy VPN OFF. Remote, VPN ON.

10. Logical traffic paths

On-site (PETCPL / wired): Clients use the VLAN gateway. Peak Energy VPN stays **OFF** so 10.80.x is not forced into the tunnel.
Remote: Peak Energy VPN **ON** → UDP 51820 → PEAK-WG. AMC peers reach CORP (.100/.101), including PeakPulse and the ACS. Admins also reach MGMT (.99) and the firewall GUI. RDP is blocked for every AMC peer except Jagadeshwar.
Keka: cin3.a.keka.com → WAN TCP 1433 → 10.80.100.10.
Guest: PETCPL-GUEST on VLAN 102, captive portal, blocked from MGMT and CORP, internet only on the weekday schedule.

Confidential — Peak Energy IT. Live PEAK-CORP-FW policy and WireGuard peer pack.